

Probability and Statistics

Least-Square Method

Dr. Yasir Ali (yali@ceme.nust.edu.pk)

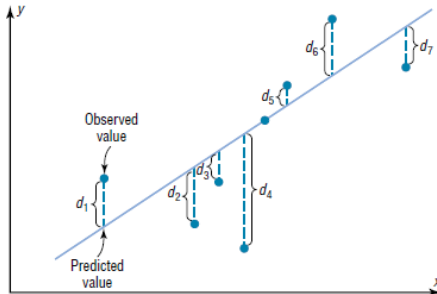
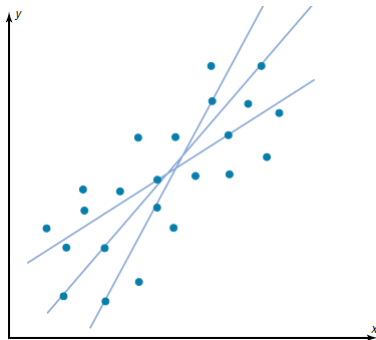
DBS&H, CEME-NUST

June 8, 2022

Line of Best Fit

Following figure shows a scatter plot for the data of two variables. It shows that several lines can be drawn on the graph near the points. Given a scatter plot, you must be able to **draw the line of best fit**.

Best fit means that the sum of the squares of the vertical distances from each point to the line is at a minimum.



Least Squares Principle

The straight line should be fitted through the given points so that the sum of the squares of the distances of those points from the straight line is minimum, where the distance is measured in the vertical direction (the y-direction)

A straight line is the simplest type of approximating curve and its equation is written as

$$Y = a + bX \quad \text{where } a \text{ and } b \text{ are the values to be determined.}$$

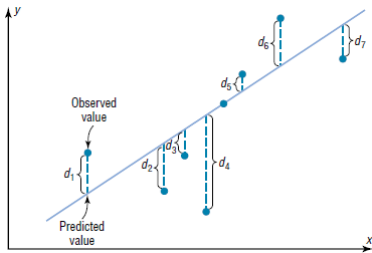
Given n pairs of observations $\{(X_i, Y_i), i = 1, 2, \dots, n\}$ to which we wish to fit a straight line. We determine the values of a and b by the *Principle of least Squares*, which calls for the minimization of the sum of the differences between the actual values of Y and the corresponding values provided by $a + bX_i$.

That is, we minimize

$$S = \sum_{i=1}^n (Y_i - a - bX_i)^2$$

To do so, we need to solve two equations:

$$\frac{\partial S}{\partial a} = 0, \quad \frac{\partial S}{\partial b} = 0$$



That is

$$\frac{\partial S}{\partial a} = 2 \sum_{i=1}^n (Y_i - a - bX_i)(-1) = 0 \text{ or } \sum_{i=1}^n Y_i - \sum_{i=1}^n a - b \sum_{i=1}^n X_i = 0$$

$$\frac{\partial S}{\partial b} = 2 \sum_{i=1}^n (Y_i - a - bX_i)(-X_i) = 0 \text{ or } \sum_{i=1}^n X_i Y_i - a \sum_{i=1}^n X_i - b \sum_{i=1}^n X_i^2 = 0$$

Now first equation gives

$$\sum_{i=1}^n Y_i - b \sum_{i=1}^n X_i = \sum_{i=1}^n a = na \quad a = \sum_{i=1}^n \frac{Y_i}{n} - b \sum_{i=1}^n \frac{X_i}{n} = \bar{Y} - b\bar{X}$$

Using this value of a in the second equation, we get

$$\sum_{i=1}^n X_i Y_i - \frac{1}{n} \left(\sum_{i=1}^n Y_i - b \sum_{i=1}^n X_i \right) \sum_{i=1}^n X_i - b \sum_{i=1}^n X_i^2 = 0$$
$$\sum_{i=1}^n X_i Y_i - \frac{1}{n} \left(\sum_{i=1}^n X_i \sum_{i=1}^n Y_i + b \left(\sum_{i=1}^n X_i \right)^2 \right) - b \sum_{i=1}^n X_i^2 = 0$$

Multiplying n on both sides we get

$$n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i + b \left(\left(\sum_{i=1}^n X_i \right)^2 - n \sum_{i=1}^n X_i^2 \right) = 0$$

Rearranging the terms we get

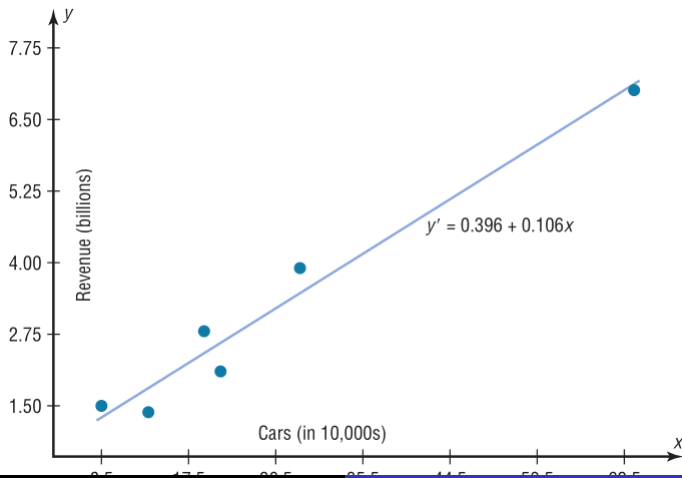
$$b = \frac{n(\sum_{i=1}^n X_i Y_i) - (\sum_{i=1}^n X_i)(\sum_{i=1}^n Y_i)}{n(\sum_{i=1}^n X_i^2) - (\sum_{i=1}^n X_i)^2}$$

and

$$a = \frac{(\sum_{i=1}^n Y_i)(\sum_{i=1}^n X_i^2) - (\sum_{i=1}^n X_i)(\sum_{i=1}^n X_i Y_i)}{n(\sum_{i=1}^n X_i^2) - (\sum_{i=1}^n X_i)^2}$$

Find the equation of the regression line for following data.

Company	A	B	C	D	E	F
Cars(thousands)	63	29	20.8	19.1	13.7	8.5
Revenue	7.0	3.9	2.1	2.8	1.4	1.5



$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$n = 6$, $\sum x = 153.8$, $\sum y = 18.7$, $\sum xy = 682.77$, and $\sum x^2 = 5859.26$

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2} = \frac{(18.7)(5859.26) - (153.8)(682.77)}{(6)(5859.26) - (153.8)^2} = 0.396$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2} = \frac{6(682.77) - (153.8)(18.7)}{(6)(5859.26) - (153.8)^2} = 0.106$$

Hence, the equation of the regression line $y = a + bx$ is

$$y = 0.396 + 0.106x$$

x	y	x^2	xy	y^2
1	1	1	1	1
3	2	9	6	4
4	4	16	16	16
6	4	36	24	16
8	5	64	40	25
9	7	81	63	49
11	8	121	88	64
14	9	196	126	81
$\Sigma x = 56$	$\Sigma y = 40$	$\Sigma x^2 = 524$	$\Sigma xy = 364$	$\Sigma y^2 = 256$

$$a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n \Sigma x^2 - (\Sigma x)^2} = \frac{(40)(524) - (56)(364)}{(8)(524) - (56)^2} = \frac{6}{11}$$

$$b = \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{n \Sigma x^2 - (\Sigma x)^2} = \frac{(8)(364) - (56)(40)}{(8)(524) - (56)^2} = \frac{7}{11}$$

The data shown for car rental companies in the United States for a recent year.

Company	Cars	Revenue
A	63.0	7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

Find the equation of the regression line for the above data

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

The data shown for car rental companies in the United States for a recent year.

Company	Cars	Revenue	xy	x^2	y^2
A	63.0	7.0	441.00	3969	49
B	29.0	3.9	113.10	841	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	2.75	72.25	2.25
$\sum x$		$\sum y$	$\sum xy$	$\sum x^2$	$\sum y^2$

Cars	Revenue	xy	x^2	y^2
63.0	7.0	441.00	3969	49
29.0	3.9	113.10	841	15.21
20.8	2.1	43.68	432.64	4.41
19.1	2.8	53.48	364.81	7.84
13.4	1.4	18.76	179.56	1.96
8.5	1.5	2.75	72.25	2.25
$\sum x = 153.8$	$\sum y = 18.7$	$\sum xy = 682.77$	$\sum x^2 = 5859.26$	$\sum y^2 = 80.67$

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2} = \frac{(18.7)(682.77) - (153.8)(682.77)}{6(5859.26) - (153.8)^2} = 0.396$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2} = \frac{6(682.77) - (153.8)(18.7)}{6(5859.26) - (153.8)^2} = 0.106$$

Hence the equation of Regression line is

$$y = 0.396 + 0.106x$$

Odd Sam found how many hours of sunshine vs how many ice creams were sold at the shop from Monday to Friday:

x	2	3	5	7	9
y	4	5	7	10	15

Fit a simple linear regression; estimate a slope and intercept.

Even Can you conclude a relationship between the class of vertebrate and whether it is endangered or threatened? Use the 0.05 level of significance.

	Mammal	Bird	Reptile	Amphibian	Fish
End	68	76	14	13	76
Thrd	13	15	23	10	61